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Modeling of H1N1 Outbreak in Rajasthan: Methods and Approaches

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Abstract

[Go to:](#)

Background:

Mathematical models could provide critical insights for informing preparedness and planning to deal with future epidemics of infectious disease.

Objective:

The study modeled the H1N1 epidemic in the city of Jaipur, Rajasthan using mathematical model for prediction of progression of epidemic and its duration.

Materials and Methods:

We iterated the model for various values of R_0 to determine the effect of variations in R_0 on the potential size and time-course of the epidemic, while keeping value of $1/\gamma$ constant. Further simulation using varying values of $1/\gamma$ were done, keeping value of R_0 constant. We attempted to fit the actual reported data and compared with prediction models.

Results:

As R_0 increases, incidence of H1N1 rises and reaches peak early. The duration of epidemic may be prolonged if R_0 is reduced. Using the parameters R_0 as 1.4 and $1/\gamma$ as 3, it estimated that there would have been 656 actually infected individuals for each reported case.

Conclusion:

The mathematical modeling can be used for predicting epidemic progression and impact of control measures. Decreasing the value of R_0 would decrease the proportion of total population infected by H1N1; however, the duration of the outbreak may be prolonged.

Keywords: H1N1, Kermack-McKendrick model, modeling

Introduction

[Go to:](#)

WHO declared H1N1 infection as a pandemic on 11 June 2009. About 208 countries reported laboratory-confirmed cases of H1N1 influenza including 12,220 deaths.⁽¹⁾ The new virus emerged

through cross-species transmission and reassortment of H1N1 antigens and recombination between swine, avian and human strains. The pandemic posed a serious threat to the world health community and was a cause of serious concerns of various governments worldwide. In India, the first laboratory-confirmed case was reported on 16 May 2009; a 23-year-old man who had travelled from United States of America to Hyderabad in South India.

The state of Rajasthan which is the largest state in India, reported its first case on 23 July 2009. The national and state governments made a serious effort to contain the spread of the disease and the resultant morbidity and mortality in the population.

The present study is based on the reported H1N1 cases in Jaipur city, the capital of the state of Rajasthan. Using the available data, an analysis of H1N1 epidemic and its patterns has been undertaken. Mathematical model has been attempted using Kermack-McKendrick model(2) for prediction of the progression of epidemic and its duration and potential of public health measures on containment and control of H1N1.

Materials and Methods

[Go to:](#)

The study was conducted for the urban areas of Jaipur city where large number of H1N1 cases and deaths have been reported. Jaipur is located south-west of Delhi and well-connected by air, road and trains with major cities in India. Jaipur is also a major tourist destination in India. The total urban population of the city is 2.3 million.(3)

The mathematical model described by Kermack and McKendrick was used for prediction of epidemic curve and number of H1N1 cases. The model is also known as the Susceptible-Infectious-Recovered (SIR) model. The model assumes that when an infectious disease strikes a community, the disease often partitions the community into three categories-individuals that are yet to be infected (susceptible people and denoted by S); infected individuals (assumed to be infectious and denoted by I); and those recovered and possess immunity to or killed by this disease (denoted by R). One infected individual is introduced into a closed population where everyone is susceptible, and each infected individual transmits influenza with probability β , to each susceptible individual they encounter. The severity of the epidemic and the initial rate of increase depend upon the value of the basic reproduction number (R_0) which is defined as an average number of new infections that one case generates, in an entirely susceptible population, during the time they are infectious. The model assumes that if $R_0 > 1$, the disease will occur in an epidemic form; however, if $R_0 < 1$, the outbreak will die out. R_0 for H1N1 influenza is equal to β -times average duration of the infectious period.

The model consists of a system of three coupled non-linear ordinary differential equations,

$$dS/dt = -\beta SI$$

$$dI/dt = \beta SI - \gamma I$$

$$dR/dt = \gamma I$$

where, β is the infection rate which determines the number of susceptible persons infected per day by the infected person, γ is the recovery rate and $1/\gamma$ is the expected infectious period or the time until recovery.

We have used the data of Mexico outbreak(4) for R_0 as 1.4 to 1.6, and for the expected infection period, $1/\gamma$, as 3 days.(5) We iterated the model for various values of $R_0 = 1.2, 1.3, 1.4, 1.5$ and 1.6 to determine the effect of variations in R_0 on the potential size and time-course of the epidemic among the population of urban Jaipur, while keeping the value of $1/\gamma$ constant at 3 days. We further simulated the model using varying values of $1/\gamma$ ranging from 2 to 6 days, while keeping the value of R_0 constant at 1.4. The mathematical models were created and run in Microsoft Office Excel 2007.

Result

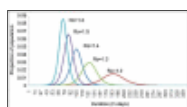
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Incidence and time distribution of H1N1

A total of 1818 cases of laboratory-confirmed H1N1 infection were detected in urban Jaipur until December 2009. The epidemic peaked in the month of November 2009 and then again in the month of December 2009, though the peak was smaller.

The simulation of the epidemic by SIR model, showed that the susceptible population decreases as the incidence (i.e., the number of individuals infected per unit time) increases. At a certain point of time, the epidemic curve reached its peak and subsequently declined, because infected individuals recover and cease to transmit the virus.

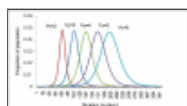
Model 1: $1/\gamma$ constant, R_0 variable Keeping $1/\gamma$ constant at 3, the model was simulated to obtain epidemic curves for varying values of $R_0 = 1.2, 1.3, 1.4, 1.5$ and 1.6 . For a value of $R_0 = 1.6$, a sharp narrow peak of epidemic curve was obtained. Decreasing the value of R_0 to 1.5 resulted in a shorter peak, although broadened the curve on the time scale. A further decrease of R_0 to 1.2 resulted in flattening of the curve. This implied that with high rate of contact, there was a rapid increase in the proportion of infected population. Decreasing the R_0 resulted in a decrease in the proportion of the infected population, but with a prolonged duration of epidemic [Figure 1].



[Figure 1](#)

Effect of change in R_0 on proportion of population infected and duration of epidemic

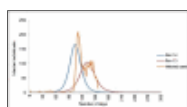
Model 2: R_0 constant, $1/\gamma$ variable Keeping R_0 constant at 1.4 , the model was simulated to obtain epidemic curves for varying values of $1/\gamma = 2, 3, 4, 5$ and 6 . For a value of $1/\gamma = 2$, a sharp narrow peak of epidemic curve was obtained. Increasing the value of $1/\gamma$ to 3 , did not change the height of peak, although resulted in a broader curve. A further increase of $1/\gamma$ to 6 , resulted in a much broader curve on the time scale with no change in the height of peak. This implied that altering the period of recovery, it did not much affect the proportion of infected population. However, it implied that with the longer periods of recovery, the duration of the epidemic was prolonged. [Figure 2].



[Figure 2](#)

Effect of change in $1/\gamma$ on proportion of population infected and duration of epidemic

For a population of urban Jaipur of 2.3 million and using the parameters R_0 as 1.4 and $1/\gamma$ as 3 , we obtained from the fitting, the estimate of 656 actually infected individuals for each reported case [Figure 3].



[Figure 3](#)

Fitting of SIR Model

Discussion

[Go to:](#)

Mathematical models are useful tools of understanding transmission dynamics and behavior of epidemics of acute infections like H1N1 and impact of potential intervention options.

The analyses in this paper were accomplished through creation of a simple but comprehensive compartment model of Kermack-McKendrick. This model offers the practical advantage of being easy to work with and amenable to solution using numerical methods. The model can be used as a tool to study the effect of public health control measures in dealing with the outbreak. The model, however, is based on assumptions that individuals mix at random and the population size remains constant over time. More complex modifications of this basic model have been developed to account for migration, seasonality and new strains arising due to cross-species transmission.

The feasibility of controlling an epidemic will critically depend on the value of the R_0 . The more severe the epidemic (i.e., the greater the value of R_0), the more intensive the interventions must be, to

significantly reduce the incidence. According to WHO, R_0 is the most amenable entity to control and therefore recommends reducing it through social distancing measures like avoiding gatherings and closing schools.⁽⁶⁾ We have modeled that decreasing the value of R_0 would decrease the proportion of total population infected by H1N1, although it might prolong the duration of the outbreak. Other studies based on modeling have also shown that behavioral interventions can be effective in mitigating the outbreak of H1N1.^(7,8)

It was a formidable challenge to develop a vaccine against the new strain of H1N1. The new vaccine could be made available in the market sometime in March 2010, till then it was only the anti-viral drugs and social and behavioral interventions that were used to manage the epidemic. Further analysis need to be made to evaluate the effectiveness of the new vaccine, and modeling the effect of mass vaccination on the future outbreaks and epidemics. Results of the study showed that if R_0 could be reduced, the epidemics can be prevented. The mathematical models could provide critical insights for informing preparedness and planning to deal with future epidemics.

Footnotes

[Go to:](#)

Source of Support: Nil

Conflict of Interest: None declared.

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[Go to:](#)

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